SPATIAL LIBRIMIX: AMBISONIC SPEECH SEPARATION DATASET TECHNICAL REPORT

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ABSTRACT

This is the technical report for Spatial Librimix, a spatial audio dataset designed for first-order ambisonics speech source separation and extraction. We present all hyperparameters and processing in detail. We provide figures representing the distributions of metadata. We introduce subsets of the dataset meant for varying difficulties of speech extraction. Design choices are motivated in the companion design paper *Spatial Analysis Of Noise Sources In Scenes Generated By The Ambisonics Image Source Model: Application To Spatial Librimix* submitted to ICASSP27. We also provide an in-depth overview of the ambisonics Image Source Model used in the design paper.

1. SPATIAL LIBRIMIX

We present Spatial Librimix (SLM), a spatial audio dataset inspired by [1, 2, 3, 4] enabling ambisonic source separation and extraction. Design choices for SLM were introduced in the design paper. This dataset contains 64.7k training, 3k validation and 6k test mixtures at 16kHZ sample rate. Mixtures are generated by summing reverberated ambisonic signals together. Each mixture includes two speech signals and a noise signal, and may include an ambient noise signal. Reverberant ambisonic signals are obtained by convolving dry signals with Spatial Room Impulse Responses (SRIRs). SRIRs are ambisonic signals that describe a scene's reverberation. In SLM, we use both synthetic and real SRIRs. Synthetic SRIRs were generated using the ambisonics Image Source Model (ISM), introduced in detail in Section 2. Real SRIRs originate from the TAU-SRIR, MOTUS and Maida Vale datasets [5, 6, 7].

1.1. Synthetic SRIRs

We generate the synthetic SRIRs using pyroomacoustics [8] Image Source Model (ISM) on simulated scenes (Eq. (20)). We generate 70.7k synthetic scenes for training (64.7k), validation (3k) and testing (3k). Each synthetic scene involves a shoebox (rectangular) room, an ideal ambisonic receiver, 3 omnidirectional point sources, 4 directional sources oriented into the corners, and a source meant to be used for late reverberation. Their parameters are randomly sampled. Scene generation parameters were chosen to represent the median parameters from a few related works [9, 10, 11, 12, 13, 14, 15, 2] as explained in the design paper. Scene generation hyperparameters are described in table 1. An example of a scene is represented in Fig 1.

RT60 values for the synthetic scenes are sampled unconventionally. Obtaining the absorption values for the walls using the inverse sabine RT60 method from pyroomacoustics applies the same absorption value to every wall. To add some variance to the wall absorptions, we randomly displace their values. Let $\tau_{60} \in [0.2, 0.8]$ be the sampled RT60 value. Let S_i be the surface of the i-th wall of the shoebox room. We define that wall's absorption value α_i as

$$\alpha_i = \frac{0.161}{6\tau_{60}S_i}. \tag{1}$$

Each α_i is additionally individually displaced by a random amount, sampled from a normal distribution $\mathcal{N}(0, \frac{1}{60}\sum_{i=1}^{6}\alpha_i)$. After displacement, absorption values are restricted to the interval $[0.01, 0.8]$. We may estimate the RT60 again using the Sabine formula

$$\hat{\tau}_{60} = \frac{0.161V}{\sum_{i=1}^{6}\alpha_i S_i}. \tag{2}$$

When generating the SRIRs, the ideal ambisonic receiver is simulated using 4 virtual microphones (Eq. (10), (21)). We use the Randomized Image Method which randomly displaces source images ($\pm$5cm) to avoid sweeping echoes from highly symmetric shoebox rooms (Eq. (6)). We finally preprocess the SRIRs with an 100Hz high-pass filter due to the ISM being particularly inaccurate at low frequencies [16]. The use of ideal ambisonic receivers for first-order ambisonics at 16kHz sampling rate is justified as real first-order-ambisonics encodings have artefacts mostly below 100Hz and above

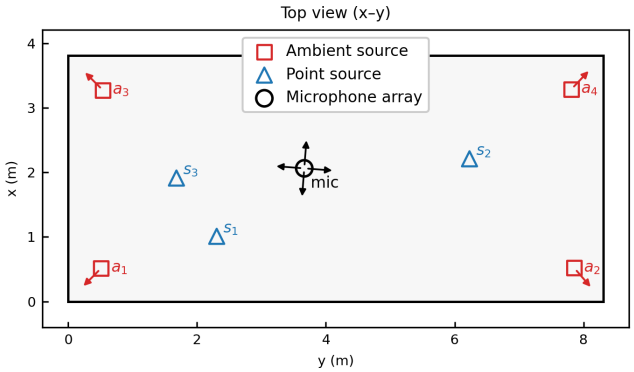


Fig. 1. Spatial Librimix scene example (*Four corners* ambient source). The ambisonic virtual microphone array is randomly rotated. Point sources are omnidirectional and randomly distributed in the shoebox room. Ambient sources are directional (hypercardioid) and oriented towards the corners.

Table 1. Synthetic scene generation hyperparameters

Parameter	**Value** or **Range**
Room width, height	$[3, 10]$m, $[2, 6]$m
Mic-to-src. distance	$[1, 5]$ m
Min. mic/src-to-wall distance	0.5m
Min. src-to-src angular difference	15°

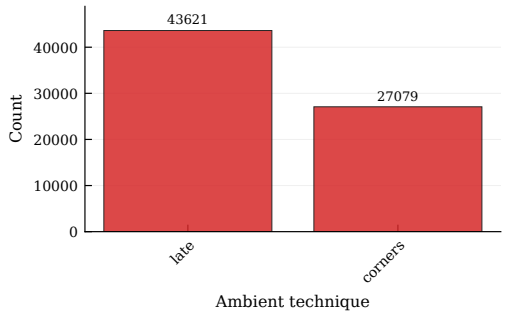


Fig. 2. Proportion of ambient noise reverberated using either the late technique or the 4 corners technique.

8000Hz [17, 18]. Ambient noise SRIRs are generated using two methods as described in the design paper. Either (1) using the late reverberation of a source randomly placed within the room or (2) summing the SRIRs from the 4 corner sources . The late reverberation is defined as the SRIR content after 0.05s. The ambient noise SRIR used for each mixture is the one with the highest diffusiveness score as introduced in the design paper. The proportion of each technique used in the dataset is represented in Fig. 2.

1.2. Real SRIRs

The TAU-SRIR, MOTUS and Maida Vale datasets [5, 6, 7] provide real SRIRs arranged in comparatively few rooms compared to what may be synthetically generated. By selecting unique sets of 3 SRIRs from each room and assigning them to the speech and noise sources, we create 3k sets of real SRIRs to be used for testing.

1.3. Mixtures

The synthetic mixtures reuse the speech and ambient noise splits from Libri2Mix [1] (LibriSpeech [19] and WHAM! Noise [20] respectively). This results in 64.7k synthetic mixtures for the train set, 3k synthetic mixtures for the validation set and 3k synthetic mixtures for the synthetic test set. We add extra noise sources from FSD50K [21] from a subset of indoor-appropriate classes as in Spatial Librispeech [2]. White sensor noise [3] may be added at runtime. For the real mixtures, we reuse the same speech and noise splits as the synthetic test set, although we notably cannot use the ambient noise sources due to not having a unique ambient SRIR.

Mixtures are created by summing reverberant speech sources first. Reverberant noise is then added with signal-to-noise ratio (SNR) sampled from $[-3 + 10\log_{10}(2), 3 + 10\log_{10}(2)]$dB. This value is chosen following the insights from the design paper, where point sources with SNR values between -3dB and 3db hinder localization relative to other point sources. The $10\log_{10}(2)$dB added is due to the two speech sources being previously mixed. Finally ambient noise is added with SNR sampled from $[-6+10\log_{10}(3), 3+10\log_{10}(3)]$, following the values from the original WHAM! paper [20].

1.4. Implementation details

This dataset was generated by first generating the metadata, and then using the metadata to generate the mixtures. This process was verified to be reproducible.

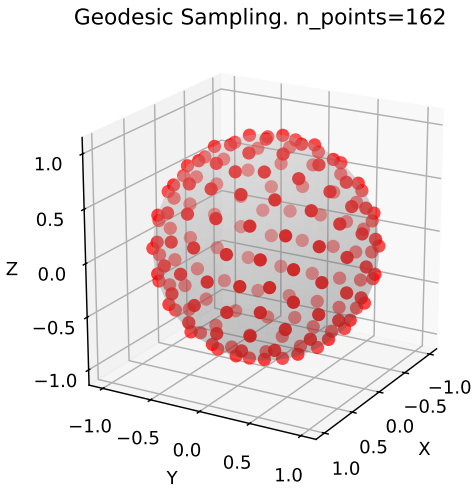


Fig. 3. Sampling used to compute diffusiveness and task metrics.

Metrics introduced in the design paper are computed and stored as metadata in a slightly different manner than in the design paper. The design paper made assumptions on the signals' self-covariance to highlight general principles. Here the metrics D (diffusiveness), SE-SNR (speech extraction SNR) and SL-SNR (speech localization SNR) are used with the actual signals from the dataset. Diffusiveness is measured on a per-source basis, as in the design paper. In the design paper, SE-SNR and SL-SNR for generic noise sources ($\diamond$N in the design paper) were measured against a generic target point source (PS in the design paper). Here, SE-SNR and SL-SNR are both measured between each source (replacing sources indexed PS in the design paper) and all other sources summed (replacing sources indexed $\diamond$N in the design paper). The metrics introduced in the design paper use maximums and means over all directions. In practice, we use a 162-point geodesic sampling to measure maximums and means, represented in Fig 3.

1.5. Options for dataset generation

Since the generation is reproducible, we provide users with additional options for personalization.

In cases where users may want their reverberant audio to be synced with the clean audio, we provide configuration flags to sync the reverberant audio with the original clean audio using GCC-PHAT [22]. This may be useful when using clean audio as a loss target.

Since ambisonic audio files are large, we only save by default the first-order ambisonics mixture, and the first-order ambisonics speech. We provide configuration flags to save reverberant pre-spatially-filtered signals using max-DI or max-RE beamformer weights, reverberant omnidirectional signals, clean signals, for any speech, noise or ambient source.

1.6. Figures

We represent the distribution of floor surfaces in Fig. 4a, room volumes in Fig. 4b, RT60 values in Fig. 4c and angular distances between speech source 1 and 2 in Fig. 4d for the 70.7k synthetic scenes generated for SLM.

For each source in SLM (Speech source 1, Speech source 2, Noise source 3, and the ambient source), we represent the distri-

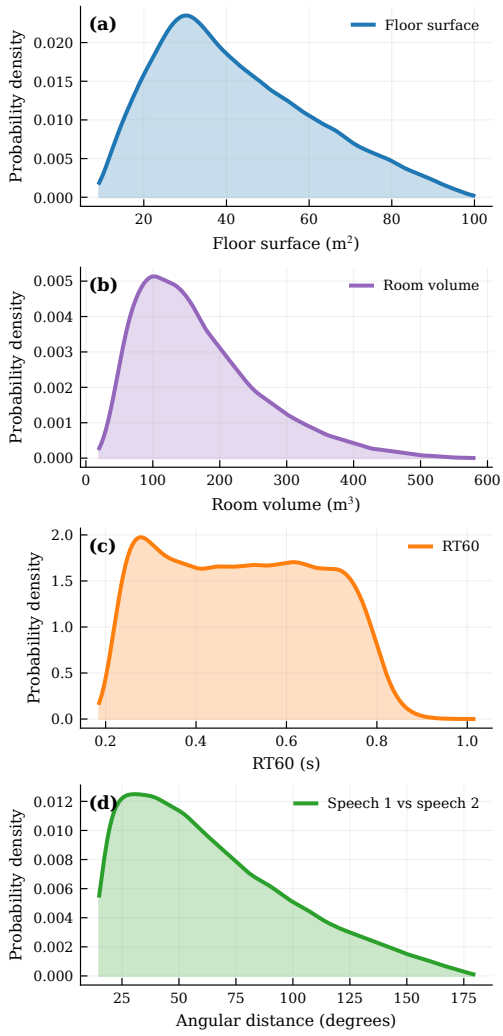


Fig. 4. Metadata distribution in Spatial Librimix.

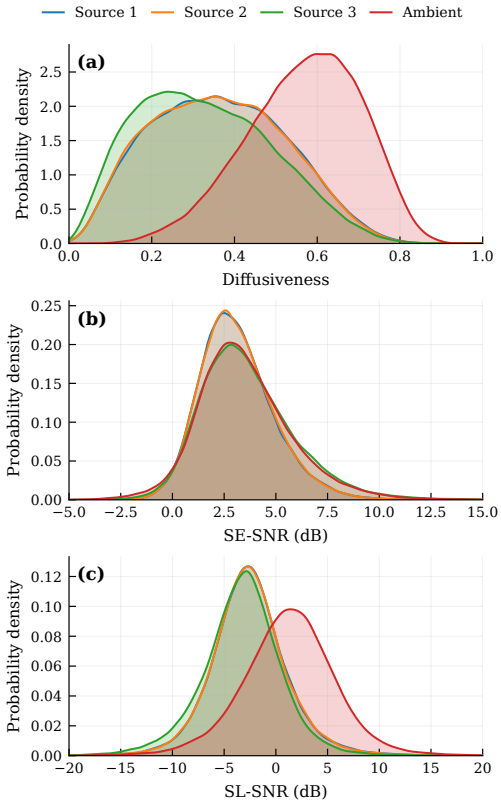


Fig. 5. Metric distribution in Spatial Librimix.

bution of Diffusiveness in Fig. 5a, Speech Extraction SNR in Fig. 5b and Source Localization SNR values in Fig. 5c for the 70.7k synthetic scenes generated for SLM.

1.7. Subsets

Using the median Speech Extraction SNR metric ($\approx$ 2.94dB), we create three subsets of the synthetic scenes generated for SLM. Scenes where both the speech sources' SE-SNR are above the median SE-SNR are included in the *Easy* subset ($\approx$ 22k scenes). Scenes where both the speech sources' SE-SNR are below the median SE-SNR are included in the *Hard* subset ($\approx$ 22k scenes). Remaining scenes are included in the *Mid* subset ($\approx$ 26k scenes). These subsets are meant to represent varying difficulties for the source extraction task. These subsets might be helpful for benchmarking source extraction models. We represent the distribution of RT60 values for the scenes in the varying subsets in Fig. 6 and the distribution of angular distances between speech source 1 and 2 for the scenes in the varying subsets in Fig. 7

2. AMBISONICS ISM

Properly annotated real spatial audio data is scarce and costly, so many spatial audio papers rely on synthetic data for their experiments. We present here in detail the Ambisonic Image Source Model (ISM) we use for reverberation in Spatial Librimix. The ISM is simple and overwhelmingly used in synthetic spatial audio experiments. Many multichannel and ambisonic speech extraction papers rely on the ISM for their experiments. It is also common for multichannel and ambisonic spatial audio datasets to rely on the ISM as their reverberation simulator. The ISM is an incomplete model of reverberation. The ISM only models specular reflections, disregarding many effects that wave-based or ray tracing solvers represent, such as diffraction. It also struggles to represent late reverberation, which it models as a massive sum of individual specular reflections. Nonetheless, it is widely used for its speed and simplicity. The ISM particularly excels at estimating the early reverberation of point sources in rectangular rooms, commonly called shoebox rooms.

Image sources The Image Source Model (ISM) models the reverberation of a source signal $\mathcal{X}_0$ as a sum of delayed and attenuated image sources signals. Let $\mathcal{O}$ be the maximum order of reflections of the ISM. The set of image source indices is

$$\mathcal{I} = \{\mathbf{i} = (i_x, i_y, i_z) \in \mathbb{Z}^3; |i_x| + |i_y| + |i_z| \leq \mathcal{O}\}. \quad (3)$$

There are $|\mathcal{I}|$ sources. We define

$$\bar{\mathbf{i}} = \mathbf{i} \bmod 2, \quad \bar{\mathbf{i}} \in \{0, 1\}^3 \quad (4)$$

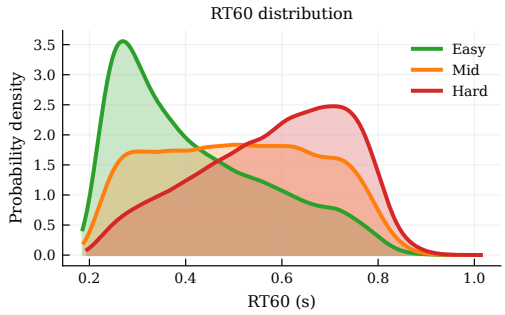


Fig. 6. RT60 values for the *Easy*, *Mid*, and *Hard* Sepeech Extraction subsets.

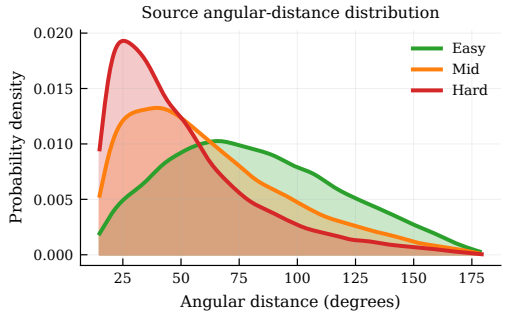


Fig. 7. Angular distances between speech source 1 and 2 for the *Easy*, *Mid*, and *Hard* Sepeech Extraction subsets.

where mod is the element-wise modulo operator. Let the room dimensions be $\mathbf{r} \in \mathbb{R}_+^3$ and the source position be $\mathbf{p} \in \mathbb{R}_+^3$ within this room such that $\mathbf{p} < \mathbf{r}$. For each image source of index $\mathbf{i} \in \mathcal{I}$, the image source position is

$$\mathbf{p_i} = (\mathbf{i} + \bar{\mathbf{i}}) \odot \mathbf{r} + (\mathbf{1}_3 - 2\,\bar{\mathbf{i}}) \odot \mathbf{s} \tag{5}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{1}_3$ is a vector of ones. The Randomized Image Source method can be used here, displacing each image source position by a small amount $\epsilon \sim \mathcal{N}(0, 2e^{-2})^3$ to avoid sweeping echoes. We write

$$\tilde{\mathbf{p}}_\mathbf{i} = \mathbf{p_i} + \epsilon. \tag{6}$$

Ideal ambisonics receiver Reverberation is defined between a source and a receiver. We define the receiver as an ideal[1] ambisonic receiver of order L. We write its position as $\mathbf{m} \in \mathbb{R}_+^3, \mathbf{m} < \mathbf{r}$. Assuming far-field conditions, the $|\mathcal{I}|$ image source signals reaching the receiver are plane-wave signals $\mathcal{X}_\mathbf{i} : \mathbb{R} \mapsto \mathbb{R}$, $\mathbf{i} \in \mathcal{I}$ with the incoming image source directions

$$\mathbf{u_i} = \frac{\tilde{\mathbf{p}}_\mathbf{i} - \mathbf{m}}{\|\tilde{\mathbf{p}}_\mathbf{i} - \mathbf{m}\|} \tag{7}$$

and azimuthal and colatitude angles

$$\phi_\mathbf{i} = \operatorname{atan2}\left(\mathrm{u}_\mathbf{i}^y, \mathrm{u}_\mathbf{i}^x\right) \in [0, 2\pi], \tag{8}$$

$$\theta_\mathbf{i} = \arccos\left(\mathrm{u}_\mathbf{i}^z\right) \in [0, \pi]. \tag{9}$$

We write $\mathbb{S}^2 = [0, 2\pi] \times [0, \pi]$. Ideal[1] ambisonic receivers are multichannel receivers, where each channel is scaled by a different real spherical harmonic $Y_l^m : \mathbb{S}^2 \mapsto \mathbb{R}$. The real spherical harmonics $Y_l^m(\phi_\mathbf{i}, \theta_\mathbf{i})$ are indexed by ambisonic order $0 \leq l \leq L$ and degree $-l \leq m \leq l$, and dependent on incoming plane-wave angles $(\phi_\mathbf{i}, \theta_\mathbf{i})$. We have[2]

$$Y_l^m(\phi_\mathbf{i}, \theta_\mathbf{i}) = N_l^{|m|} P_l^{|m|}(\cos(\theta_\mathbf{i}))\Phi_m, \tag{10}$$

$$\Phi_m = \begin{cases} \sqrt{2}\sin(|m|\phi_\mathbf{i}) & \text{if } m < 0 \\ 1 & \text{if } m = 0 \\ \sqrt{2}\cos(m\phi_\mathbf{i}) & \text{if } m > 0 \end{cases}, \tag{11}$$

$$N_l^{|m|} = \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}, \tag{12}$$

where P_l^m are the associated Legendre polynomials.

Wall attenuation Sound reflecting off a wall loses a part of its energy due to wall absorption. Let $\boldsymbol{\alpha} \in [0, 1]^6$ be the absorption coefficients[3] of the 6 walls of a shoebox room. Let the transmission coefficient of each wall be

$$\mathbf{t} = \sqrt{1 - \boldsymbol{\alpha}}. \tag{13}$$

For each image source index $\mathbf{i}$, we define the number of reflections on "low" walls and "high" walls as

$$\mathbf{e}_\mathbf{i}^{\text{lo}} = \frac{\mathbf{i} - \bar{\mathbf{i}}}{2}, \quad \mathbf{e}_\mathbf{i}^{\text{hi}} = \frac{\mathbf{i} + \bar{\mathbf{i}}}{2}. \tag{14}$$

Each image source's final wall attenuation is a product of the wall transmission coefficients of all the walls "traversed". The walls "traversed" are on the path between the image source $\mathbf{p_i}$ and the receiver $\mathbf{m}$. The closed form is

$$A_\mathbf{i}^{\text{walls}} = t_{\text{front}}^{|\mathbf{e}_{ix}^{\text{lo}}|}\, t_{\text{back}}^{|\mathbf{e}_{ix}^{\text{hi}}|}\, t_{\text{left}}^{|\mathbf{e}_{iy}^{\text{lo}}|}\, t_{\text{right}}^{|\mathbf{e}_{iy}^{\text{hi}}|}\, t_{\text{up}}^{|\mathbf{e}_{iz}^{\text{lo}}|}\, t_{\text{down}}^{|\mathbf{e}_{iz}^{\text{hi}}|}. \tag{15}$$

Flying distance attenuation Image source signals are attenuated inversely proportional to image source distance:

$$A_\mathbf{i}^{\text{dist}} = \frac{1}{\|\tilde{\mathbf{p}}_\mathbf{i} - \mathbf{m}\|}. \tag{16}$$

Flying time delay Each image source's signal $\mathcal{X}_\mathbf{i}$ is delayed by a time $\tau_\mathbf{i}$ relative to the original source's signal $\mathcal{X}_0$. Let

$$\tau_\mathbf{i} = \frac{\|\tilde{\mathbf{p}}_\mathbf{i} - \mathbf{m}\|}{c} \tag{17}$$

with c the speed of sound in m/s. We have $\mathcal{X}_\mathbf{i}(t) = \mathcal{X}_0(t + \tau_\mathbf{i})$.

[1] We use ambisonics as an array-agnostic format for spatial audio. In practice, ambisonic signals are computed from a Spherical Microphone Array (SMA) and converted to ambisonics using artefact-prone encoding. These artefacts aren't explored in this paper.

[2] With standard SN3D normalization, without Condon–Shortley phase, following the formalism from [23, p. 176,187].

[3] Defining frequency-dependent absorption coefficients ($\alpha(f)$) is possible. In the case of an ideal ambisonic receiver, this simply leads to a frequency-dependent definition of $A_\mathbf{i}^{\text{walls}}$ in Eq. (15). In the real case, spherical microphone arrays converted to ambisonics have a order-specific frequency gain and phase response. This is negligeable for 16kHz 1st-order ambisonics.

Directional source attenuation Directional sources emanate sound in direction $\mathbf{u}_{\rightarrow} \in \mathbb{R}^3, \|\mathbf{u}_{\rightarrow}\| = 1$. Let $\mathbf{u}_{\mathbf{i}\rightarrow}$ be the emanating image source direction, defined as

$$\mathbf{u}_{\mathbf{i}\rightarrow} = (\mathbf{1} - 2\,\bar{\mathbf{i}}) \odot \mathbf{u}_{\rightarrow}. \tag{18}$$

Assuming the directional source has a cardioid directivity, the directional attenuation of image source $\mathbf{i}$ is

$$A_{\mathbf{i}}^{\text{dir}} = \frac{1 - \mathbf{u}_{\mathbf{i}\rightarrow}^{\top}\mathbf{u}_{\mathbf{i}}}{2}. \tag{19}$$

If a source isn't directional, then this factor may be omitted.

ISM-reverberated signal Aggregating all image source signals $\mathbf{i} \in \mathcal{I}$, an ISM-reverberated signal reaching the ideal ambisonic microphone of order L is

$$\mathbf{X}_L(t) = \sum_{\mathbf{i}\in\mathcal{I}} A_{\mathbf{i}}^{\text{walls}} A_{\mathbf{i}}^{\text{dist}} A_{\mathbf{i}}^{\text{dir}} \mathcal{X}_0(t + \tau_{\mathbf{i}}) \boldsymbol{y}_L(\phi_{\mathbf{i}}, \theta_{\mathbf{i}}) \tag{20}$$

where $\boldsymbol{y}_L : S^2 \mapsto \mathbb{R}^{(L+1)^2}$ is the L-order vector of real spherical harmonics

$$\boldsymbol{y}_L(\phi_{\mathbf{i}}, \theta_{\mathbf{i}}) = [Y_l^m(\phi_{\mathbf{i}}, \theta_{\mathbf{i}})]_{l\in\{0,\,\ldots,\,L\}}^{-l\le m\le l}. \tag{21}$$

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